Support Vector Machines

Yu Sheng

Center for Statistical Science

Formulation

- Our training data consists of N pairs $(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$, with $x_i \in \mathbb{R}^p$ and $y_i \in \{-1, 1\}$.
- Define a hyperplane by

$$L = \{x : x^T \beta + \beta_0 = 0\},$$

which we will use as a classification boundary.

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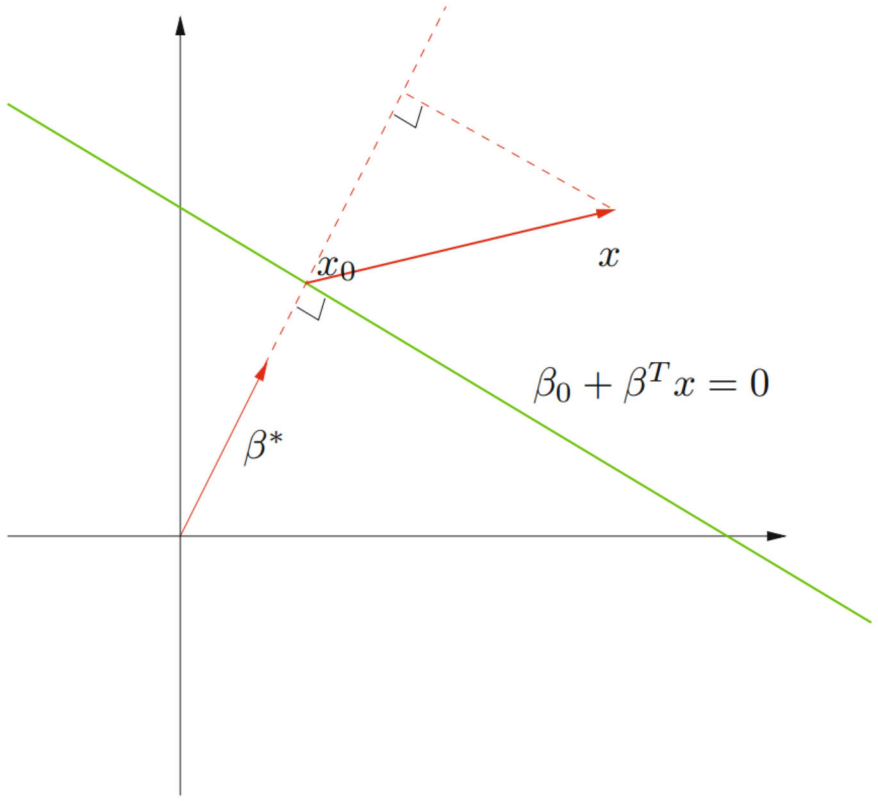
Some Vector Algebra



- For any point x_0 in L, $x_0^T \beta = -\beta_0$.

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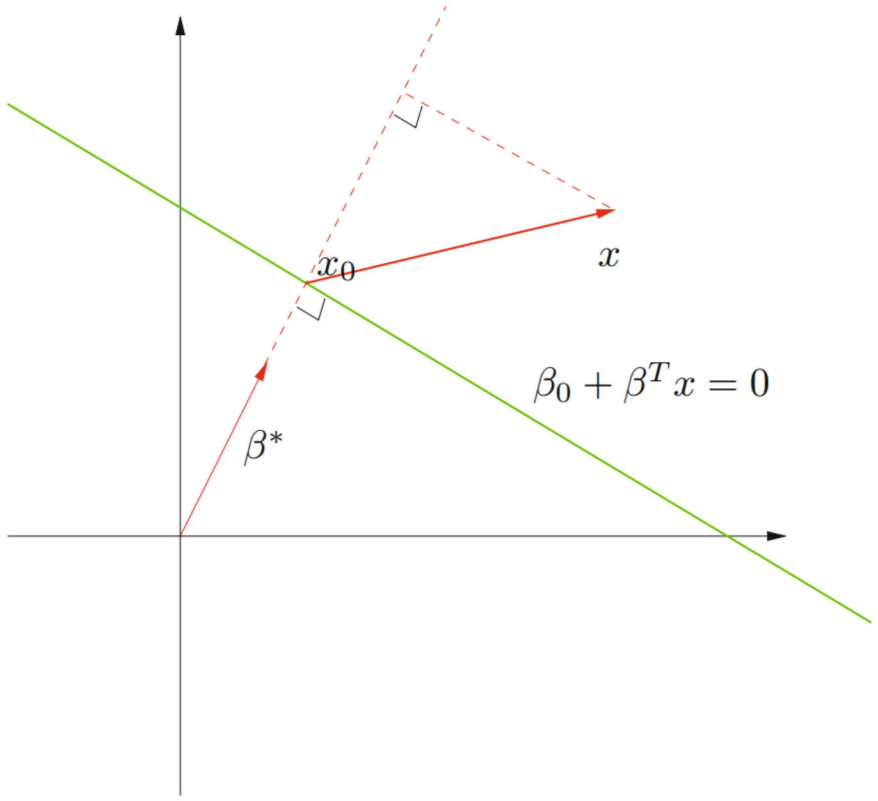
Some Vector Algebra



- For any two points x_1 and x_2 lying in L, $(x_1 - x_2)^T \beta = 0$, and hence $\beta^* = \beta / \|\beta\|$ is the unit vector orthogonal to the surface of L.

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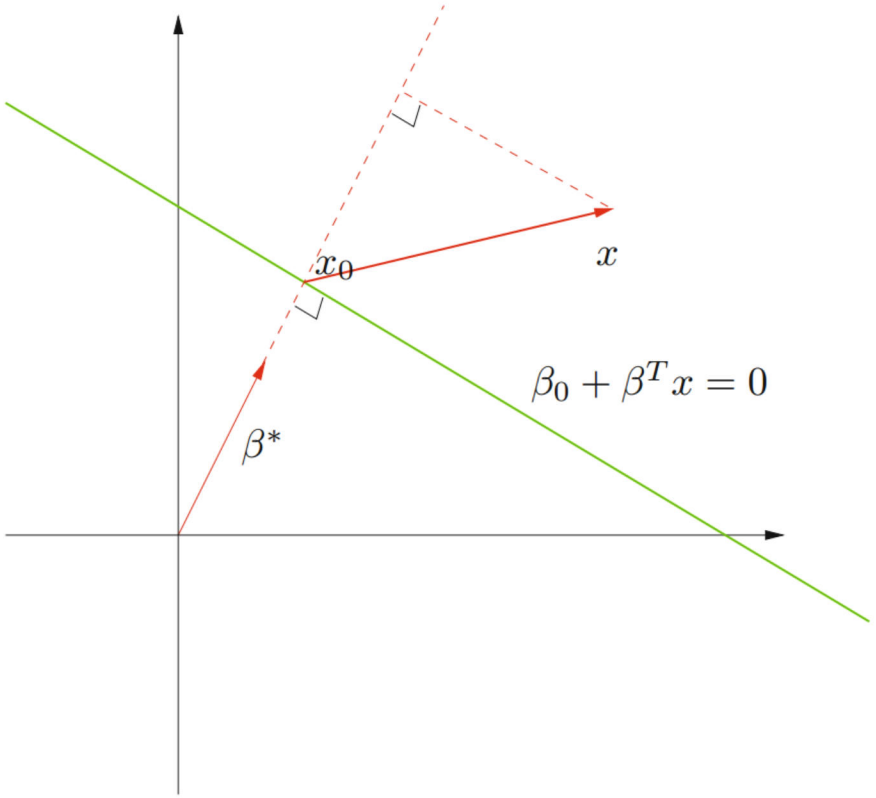
Some Vector Algebra



- The signed distance of any point x to L is given by

$$(x - x_0)^T \beta^* = \frac{1}{\|\beta\|}(x^T\beta + \beta_0).$$

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Classification Rule

- For a point x "above" the hyperplane L (along the direction of β),

$$x^T\beta + \beta_0 > 0.$$

- For x "below" L ,

$$x^T\beta + \beta_0 < 0.$$

- Thus, it is natural to classify by

$$\hat{y}(x) = \text{sign}\left(x^T\beta + \beta_0\right).$$

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Optimal Separating Hyperplane

- Assume that the two classes in the training data are perfectly separable by the hyperplane

$$L = \{x: x^T\beta + \beta_0 = 0\},$$

then for any (x_i, y_i), we have $y_i(x_i{}^T\beta + \beta_0) > 0$.

- $\frac{1}{\|\beta\|} y_i(x_i{}^T\beta + \beta_0)$ is the distance of x_i to L.
- We can define optimality by maximizing the distance to all the points.

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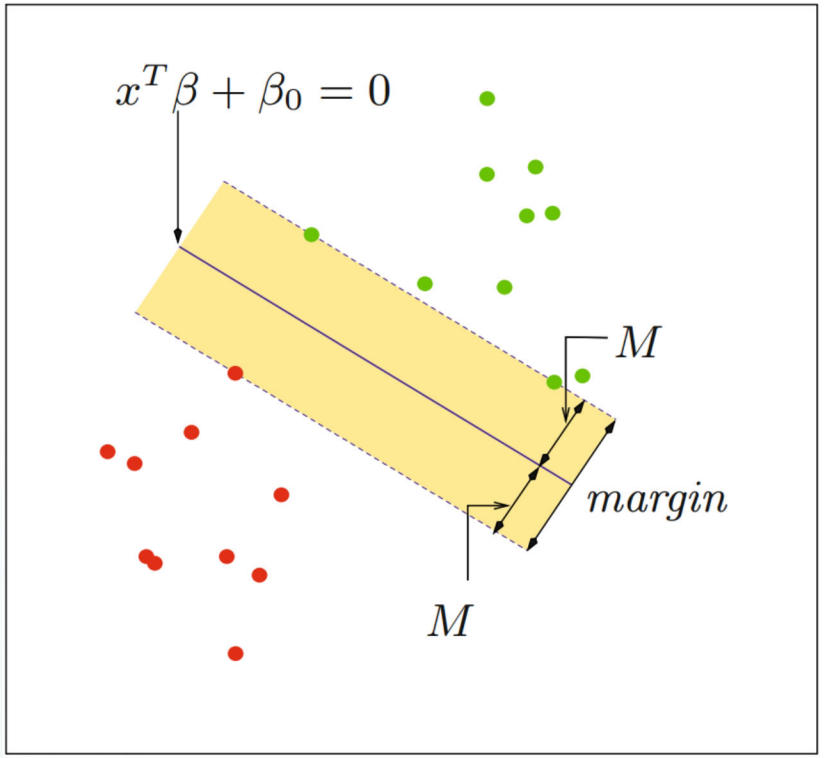
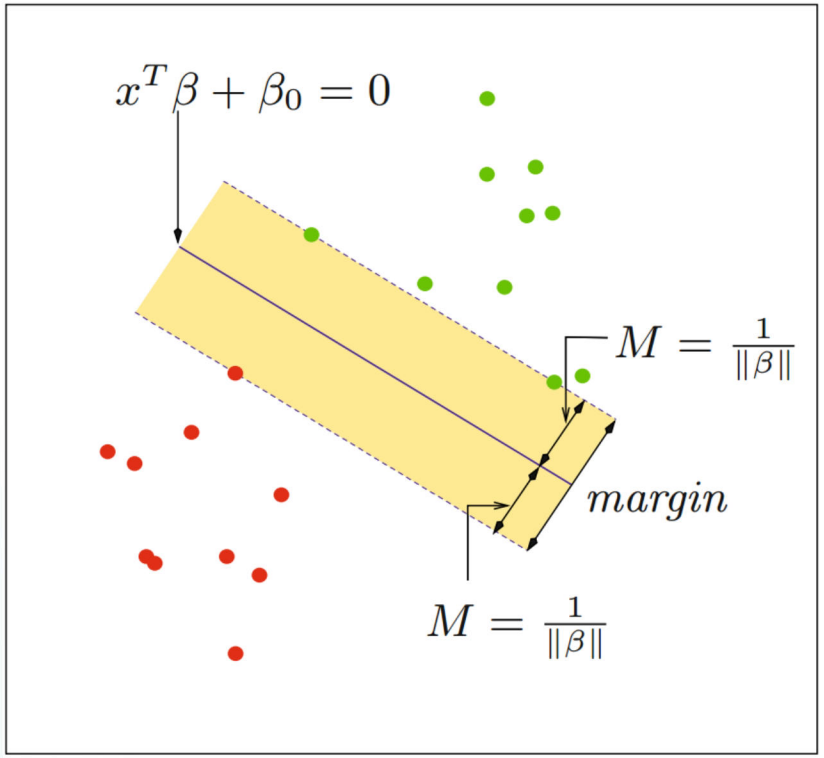
Optimal Separating Hyperplane



$$\max_{\beta,\beta_0,M} \quad M$$

$$\text{s.t.} \quad \frac{1}{\|\beta\|} y_i\left(x_i{}^T\beta + \beta_0\right) \geq M, \text{for } i = 1, \ldots, N$$

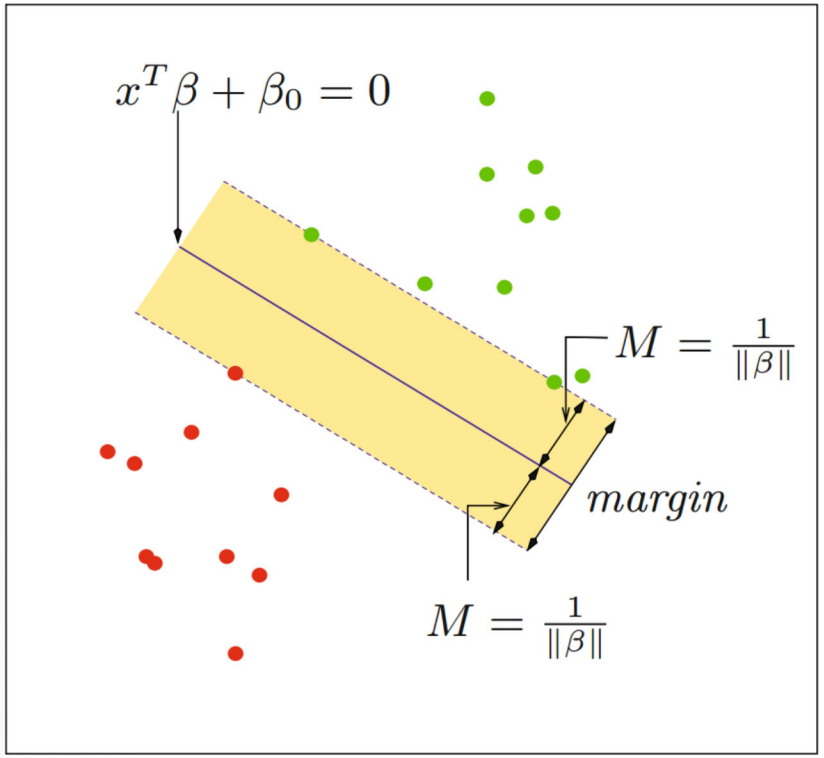
Optimal Separating Hyperplane



$$\max_{\beta,\beta_0} \frac{1}{\|\beta\|}$$

$$\text{s.t. } \frac{1}{\|\beta\|} y_i\left(x_i{}^T\beta + \beta_0\right) \geq \frac{1}{\|\beta\|}, \text{for } i = 1, \ldots, N$$

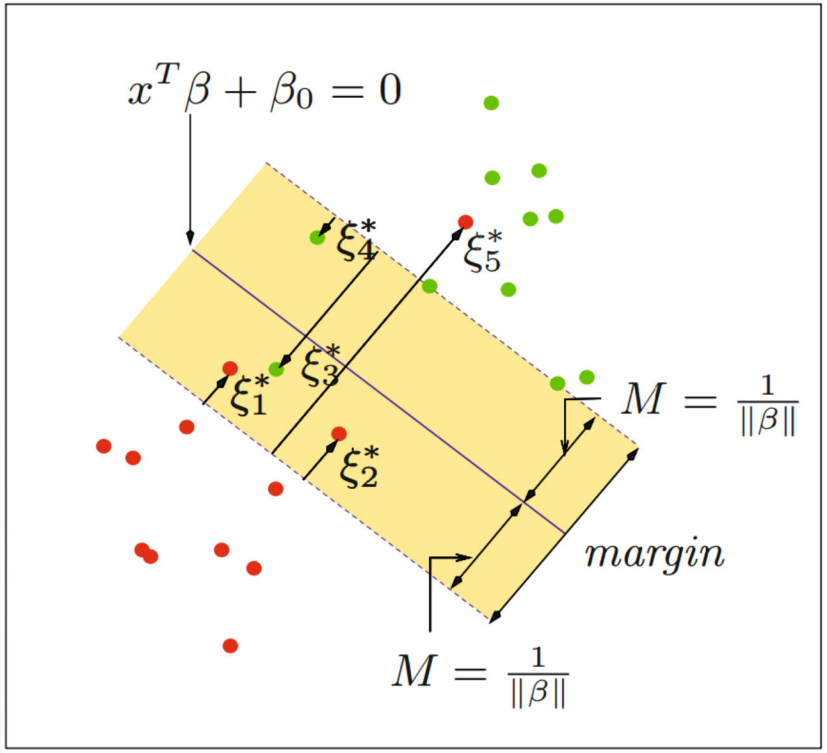
Optimal Separating Hyperplane



$$\min_{\beta,\beta_0} \|\beta\|$$

$$\text{s.t. } y_i(x_i{}^T\beta + \beta_0) \geq 1, \text{for } i = 1, \ldots, N$$

Non-separable Cases

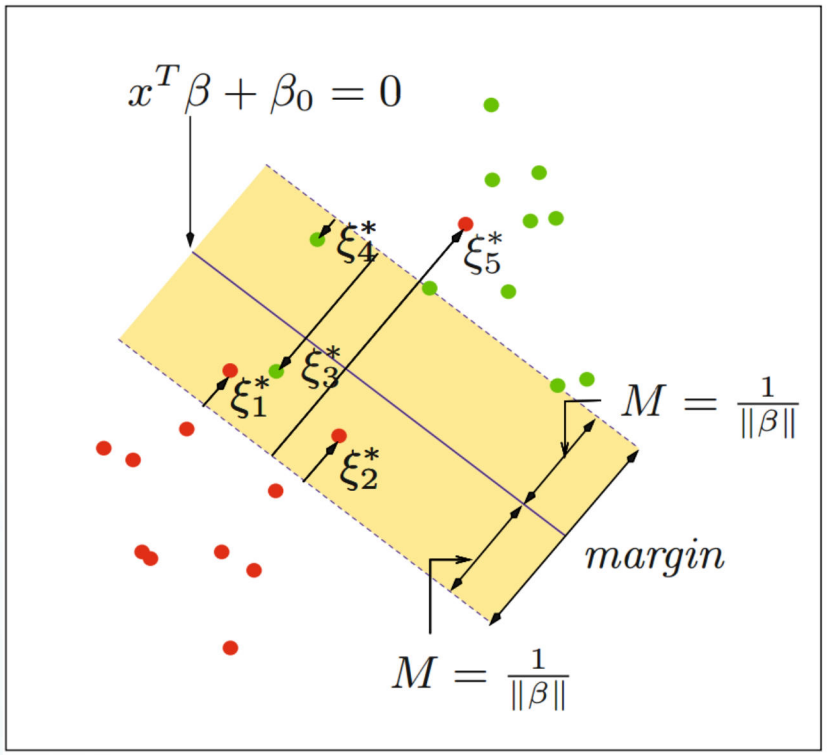


$$\max_{\beta,\beta_0,M,\xi_i} M$$

$$\text{s.t. } y_i\left(x_i{}^T\beta + \beta_0\right)/\|\beta\| \geq M(1-\xi_i), \quad \text{for } i = 1,\ldots,N$$

$$\sum_{i=1}^{N}\xi_i \leq \text{constant}, \quad \xi_i \geq 0, \quad \text{for } i = 1,\ldots,N$$

Non-separable Cases

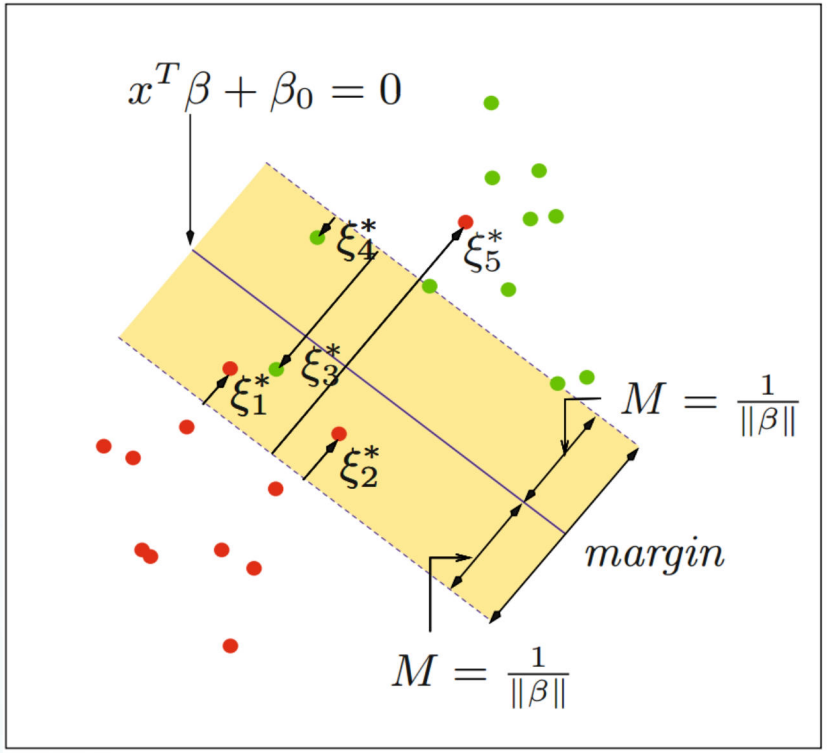


$$\min_{\beta,\beta_0,\xi_i} \|\beta\|$$

$$\text{s.t. } y_i\left(x_i{}^T\beta + \beta_0\right) \geq 1 - \xi_i, \quad \text{for } i = 1, \ldots, N$$

$$\sum_{i=1}^{N} \xi_i \leq \text{constant}, \quad \xi_i \geq 0, \quad \text{for } i = 1, \ldots, N$$

Non-separable Cases

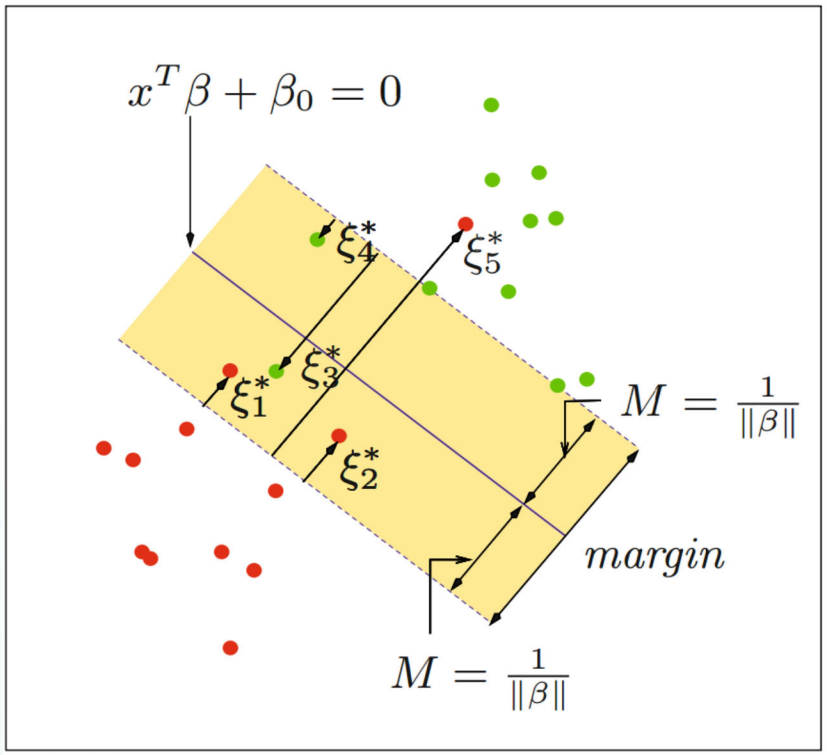


$$\min_{\beta,\beta_0,\xi_i} \frac{1}{2}\|\beta\|^2$$

$$\text{s.t. } y_i\left(x_i{}^T\beta + \beta_0\right) \geq 1 - \xi_i, \quad \text{for } i = 1, \ldots, N$$

$$\sum_{i=1}^{N} \xi_i \leq \text{constant}, \ \xi_i \geq 0, \ \text{for } i = 1, \ldots, N$$

Non-separable Cases



$$\min_{\beta,\beta_0,\xi_i} \ \frac{1}{2}\|\beta\|^2 + C\sum_{i=1}^{N}\xi_i$$

$$\text{s.t. } y_i\left(x_i{}^T\beta + \beta_0\right) \geq 1 - \xi_i \quad \text{, for } i = 1, \ldots, N$$

$$\xi_i \geq 0 \quad \text{, for } i = 1, \ldots, N$$

Lagrangian Duality

- Consider the following nonlinear programming Problem P, which we call the *primal problem*.

Primal Problem P

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0 \text{ for } i = 1, \ldots, m \\ & h_i(x) = 0 \text{ for } i = 1, \ldots, l \\ & x \in X. \end{aligned}$$

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Lagrangian Duality

- **Lagrangian Dual Problem D**

$$\max_{u,v} \quad \theta(u,v)$$

$$\text{s.t.} \quad u_i \geq 0 \text{ for } i = 1, \dots m,$$

where

$$\theta(u,v) = \inf\left\{f(x) + \sum_{i=1}^{m} u_i g_i(x) + \sum_{i=1}^{l} v_i h_i(x) : x \in X\right\}.$$

- Note that X is not required to be open as in the KKT conditions. This allows us to move some of the constrains to X, which can be convenient.

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Quiz

- Primal:

$$\min \ x_1^2 + x_2^2$$

$$\text{s.t.} \ x_1 + x_2 \geq 4$$

$$x_1, x_2 \geq 0$$

Let $X = \{(x_1, x_2): x_1, x_2 \geq 0\}$. Write out the dual problem.

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Lagrangian Duality

- **Primal Problem P**

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & g(x) \le 0 \\ & h(x) = 0 \\ & x \in X. \end{aligned}$$

- **Lagrangian Dual Problem D**

$$\begin{aligned} \max_{u,v} \quad & \theta(u,v) \\ \text{s.t.} \quad & u \ge 0 \end{aligned}$$

where $\theta(u,v) = \inf\{f(x) + u^T g(x) + v^T h(x) : x \in X\}$.

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Lagrangian Duality

- **Weak Duality Theorem**

  Let x be a feasible solution to Problem P; that is, $x \in X$, $g(x) \leq 0$, and $h(x) = 0$. Also, let (u, v) be a feasible solution to Problem D; that is $u \geq 0$.Then $f(x) \geq \theta(u, v)$.

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Lagrangian Duality

- **Corollary 1**

  $\inf\{f(x): x \in X, g(x) \leq 0, h(x) = 0\} \geq \sup\{\theta(u, v): u \geq 0\}$.

- **Corollary 2**

  If $f(\bar{x}) = \theta(\bar{u}, \bar{v})$, where $\bar{u} \geq 0$ and $\bar{x} \in \{x \in X, g(x) \leq 0, h(x) = 0\}$, then $\bar{x}$ and $(\bar{u}, \bar{v})$ solve the primal and dual problems, respectively.

- **Corollary 3**

  If $\inf\{f(x): x \in X, g(x) \leq 0, h(x) = 0\} = -\infty$, then $\theta(u, v) = -\infty$ for each $u \geq 0$.

- **Corollary 4**

  If $\sup\{\theta(u, v): u \geq 0\} = \infty$, then the primal problem has no feasible solution.

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Lagrangian Duality

- **Strong Duality Theorem**

Let X be a nonempty convex set in $\mathbb{R}^n$, let $f: \mathbb{R}^n \to \mathbb{R}$ and $g: \mathbb{R}^n \to \mathbb{R}^m$ be convex, and let $h: \mathbb{R}^n \to \mathbb{R}^l$ be affine (i.e., $h(x) = Ax - b$). Suppose that the following constraints hold true: There exists an $\hat{x} \in X$ such that $g(\hat{x}) < 0$ and $h(\hat{x}) = 0$, and $0 \in \text{int}\, h(X)$, where $h(X) = \{h(x): x \in X\}$. Then

$$\inf\{f(x): x \in X, g(x) \leq 0, h(x) = 0\} = \sup\{\theta(u, v): u \geq 0\}.$$

Furthermore, if the inf is finite and achieved at $\bar{x}$, then $\sup\{\theta(u, v): u \geq 0\}$ is achieved at some point $(\bar{u}, \bar{v})$, $\bar{u} \geq 0$, with $\bar{u}^T g(\bar{x}) = 0$.

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Lagrangian Duality

The logic behind

$$\begin{aligned} &\min f(x) \\ &\text{s.t. } g(x) \le c \end{aligned} \quad \Leftrightarrow \quad \min f(x) + \lambda g(x)$$

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Optimal Separating Hyperplane

- Primal:

$$\min_{\beta,\beta_0,\xi_i} \ \frac{1}{2}\|\beta\|^2 + C\sum_{i=1}^{N}\xi_i$$

$$\text{s.t. } y_i\left(x_i{}^T\beta + \beta_0\right) \geq 1 - \xi_i \quad \text{, for } i = 1, \dots, N$$

$$\xi_i \geq 0 \quad \text{, for } i = 1, \dots, N$$

- Lagrange Primal:

$$L_p = \frac{1}{2}\|\beta\|^2 + C\sum_{i=1}^{N}\xi_i - \sum_{i=1}^{N}\alpha_i[y_i\left(x_i{}^T\beta + \beta_0\right) - (1 - \xi_i)] - \sum_{i=1}^{N}\mu_i\xi_i$$

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Optimal Separating Hyperplane

- Setting the derivatives to zero, we get

$$\frac{\partial L_p}{\partial \beta} : \beta = \sum_{i=1}^{N} \alpha_i y_i x_i$$

$$\frac{\partial L_p}{\partial \beta_0} : \sum_{i=1}^{N} \alpha_i y_i = 0$$

$$\frac{\partial L_p}{\partial \xi_i} : \alpha_i = C - \mu_i$$

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Optimal Separating Hyperplane

- Substituting into the Lagrange primal, we obtain the Lagrange dual

$$L_D = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

- We maximize L_D subject to

$$0 \leq \alpha_i \leq C$$

and

$$\sum_{i=1}^{N} \alpha_i y_i = 0$$

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Support Vectors

- The KKT necessary conditions include

$$\hat{\alpha}_i\left[y_i\left(x_i{}^T\hat{\beta}+\hat{\beta}_0\right)-\left(1-\hat{\xi}_i\right)\right]=0, \text{for i}=1,\ldots,\text{N}$$

$$\mu_i\hat{\xi}_i=0, \text{for } i=1,\ldots,N$$

- These imply

$$y_i\hat{f}(x_i)>1\Rightarrow\hat{\alpha}_i=0$$

$$y_i\hat{f}(x_i)<1\Rightarrow\hat{\alpha}_i=C$$

$$y_i\hat{f}(x_i)=1\Rightarrow 0\leq\hat{\alpha}_i\leq C$$

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Support Vectors

- Solution:

$$\hat{\beta} = \sum_{i=1}^{N} \hat{\alpha}_i y_i x_i = \sum_{i \in S} \hat{\alpha}_i y_i x_i$$

where $S = \{i: \hat{\alpha}_i > 0\}$. Samples in S are called *support vectors*.

$$\hat{\beta}_0 = y_{i^*} - x_{i^*}^T \hat{\beta}$$

where i^* is a margin point ($\hat{\alpha}_{i^*} > 0, \hat{\xi}_{i^*} = 0$).

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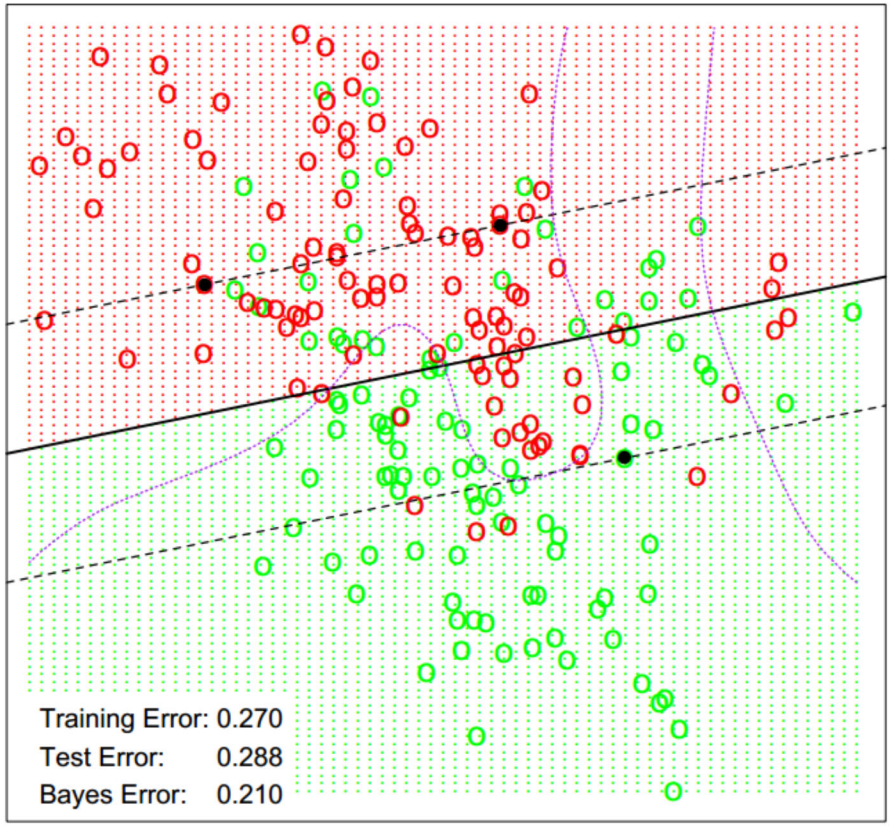
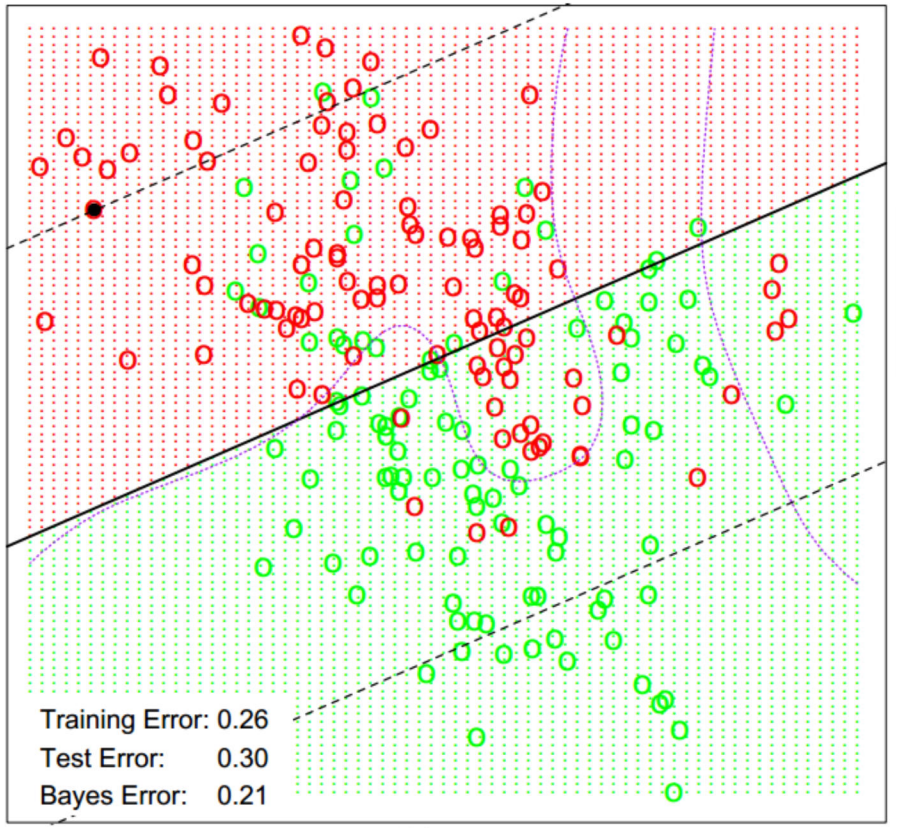
Optimal Separating Hyperplane



$C = 10000$



$C = 0.01$

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Feature Expansion

- Enlarge the input space via basis expansion ($p \to q$):

$$h(x) = \left(h_1(x), h_2(x), \ldots, h_q(x)\right)^T$$

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Feature Expansion

- Example: 2nd degree polynomial in $\mathbb{R}^2$. Choose:

$$h_1(x) = 1$$
$$h_2(x) = \sqrt{2}x_1$$
$$h_3(x) = \sqrt{2}x_2$$
$$h_4(x) = x_1^2$$
$$h_5(x) = x_2^2$$
$$h_6(x) = \sqrt{2}x_1x_2$$

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Feature Expansion

- Lagrange dual and solution become

$$L_D = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j \langle h(x_i), h(x_j) \rangle$$

and

$$\hat{y}(x) = \sum_{i \in S} \hat{\alpha}_i y_i \langle h(x), h(x_i) \rangle + \hat{\beta}_0$$

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Kernel Trick

- L_D and $\hat{y}(x)$ involve $h(x)$ only through inner-products

$$K(x, x') = \langle h(x), h(x') \rangle$$

Given a suitable kernel function $K(x, x')$, we don't need $h(x)$ at all.

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Kernel Trick

- Example Continued:

  If we choose

$$K(x, x') = (1 + \langle x, x' \rangle)^2$$

  Then

$$K(x, x') = (1 + x_1 x_1' + x_2 x_2')^2 = \langle h(x), h(x') \rangle$$

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Popular Kernels

- d-th degree polynomial: $K(x, x') = (1 + \langle x, x' \rangle)^d$
  - C_{p+d}^{d} features
- radial basis: $K(x, x') = \exp\left(-\|x - x'\|^2 / \sigma^2\right)$
  - theoretically infinite number of features

- What kind of functions can be kernels?

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Mercer's Theorem

- A symmetric function $K(x, x')$ can be expressed as an inner product

$$K(x, x') = \langle h(x), h(x') \rangle$$

for some h if and only if $K(x, x')$ is positive semidefinite, i.e.,

$$\int K(x, x') g(x) g(x') \, \mathrm{d}x \, \mathrm{d}x' \geq 0, \forall g$$

or equivalently, for any $n \in \mathbb{Z}^+$, and any set of real numbers $\{a_1, a_2, \ldots, a_n\}$ and vectors $x_1, x_2, \ldots, x_n$, we have $\sum_{i,j=1}^{n} a_i a_j K(x_i, x_j) \geq 0$.

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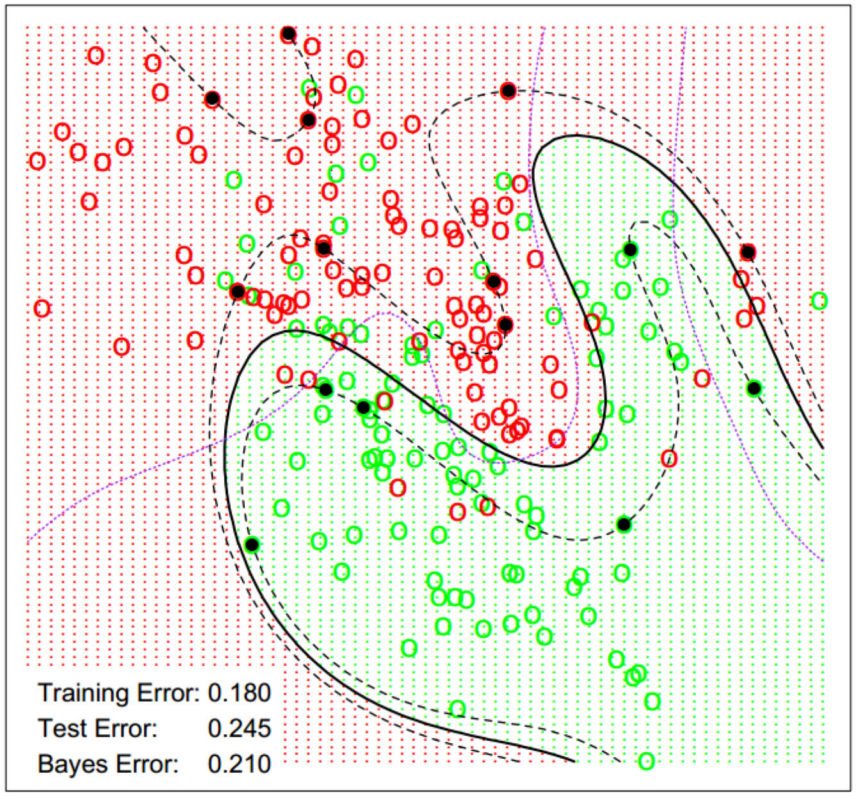
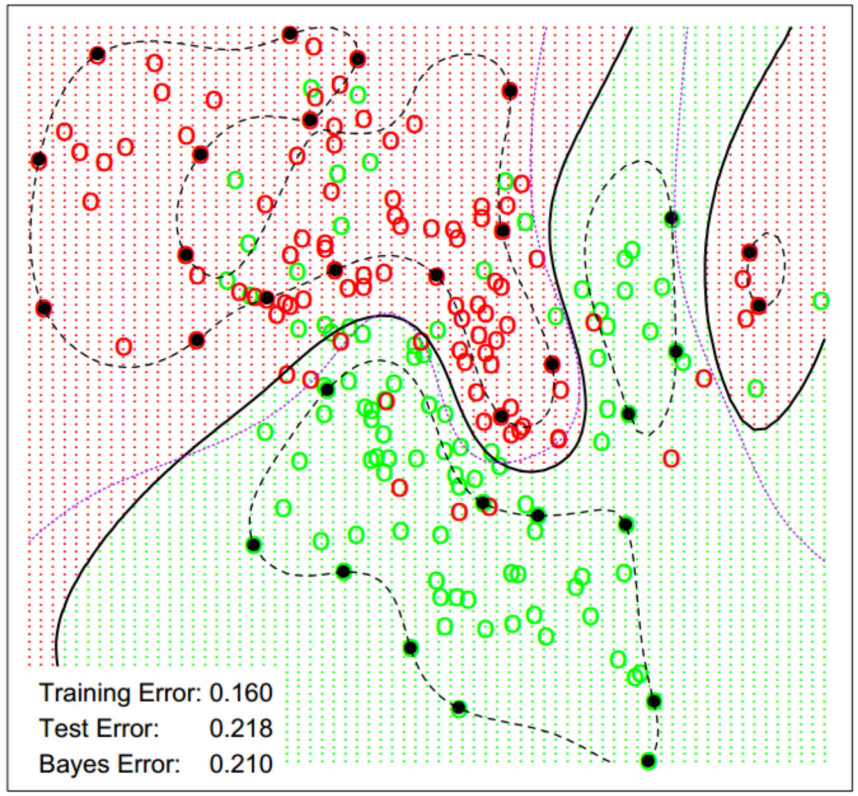
Nonlinear SVMs





- Quiz: Why does RBF kernel has better performance?

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Kernel Engineering

- The kernel function represents similarity.

$$\hat{y}(x) = h(x)^T \hat{\beta} + \hat{\beta}_0$$
$$= \sum_{i \in S} \hat{\alpha}_i y_i \langle h(x), h(x_i) \rangle + \hat{\beta}_0$$

- Designing features becomes designing kernels.
  - String kernels (Text similarity, gene analysis)
  - Tree kernels (NLP):

    https://en.wikipedia.org/wiki/Tree_kernel
  - Graph kernels (Bio/Chemoinformatic, social networks)

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Example: Substring kernel

	c-a	c-t	a-t	b-a	b-t	c-r	a-r	b-r
$\phi(\text{cat})$	λ^2	λ^3	λ^2	0	0	0	0	0
$\phi(\text{car})$	λ^2	0	0	0	0	λ^3	λ^2	0
$\phi(\text{bat})$	0	0	λ^2	λ^2	λ^3	0	0	0
$\phi(\text{bar})$	0	0	0	λ^2	0	0	λ^2	λ^3

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Example: Substring kernel

Definition 1 (String subsequence kernel- SSK) *Let Σ be a finite alphabet. A string is a finite sequence of characters from Σ, including the empty sequence. For strings s, t, we denote by $|s|$ the length of the string $s = s_1 \dots s_{|s|}$, and by st the string obtained by concatenating the strings s and t. The string $s[i:j]$ is the substring $s_i \dots s_j$ of s. We say that u is a subsequence of s, if there exist indices $\mathbf{i} = (i_1, \dots, i_{|u|})$, with $1 \le i_1 < \dots < i_{|u|} \le |s|$, such that $u_j = s_{i_j}$, for $j = 1, \dots, |u|$, or $u = s[\mathbf{i}]$ for short. The length $l(\mathbf{i})$ of the subsequence in s is $i_{|u|} - i_1 + 1$. We denote by Σ^n the set of all finite strings of length n, and by Σ^* the set of all strings*

$$\Sigma^* = \bigcup_{n=0}^{\infty} \Sigma^n. \tag{1}$$

We now define feature spaces $F_n = \mathbb{R}^{\Sigma^n}$. The feature mapping ϕ for a string s is given by defining the u coordinate $\phi_u(s)$ for each $u \in \Sigma^n$. We define

$$\phi_u(s) = \sum_{\mathbf{i}: u = s[\mathbf{i}]} \lambda^{l(\mathbf{i})}, \tag{2}$$

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Example: Substring kernel

for some $\lambda \leq 1$. These features measure the number of occurrences of subsequences in the string s weighting them according to their lengths. Hence, the inner product of the feature vectors for two strings s and t give a sum over all common subsequences weighted according to their frequency of occurrence and lengths

$$\begin{aligned} K_n(s,t) &= \sum_{u \in \Sigma^n} \langle \phi_u(s) \cdot \phi_u(t) \rangle = \sum_{u \in \Sigma^n} \sum_{\mathbf{i}: u = s[\mathbf{i}]} \lambda^{l(\mathbf{i})} \sum_{\mathbf{j}: u = t[\mathbf{j}]} \lambda^{l(\mathbf{j})} \\ &= \sum_{u \in \Sigma^n} \sum_{\mathbf{i}: u = s[\mathbf{i}]} \sum_{\mathbf{j}: u = t[\mathbf{j}]} \lambda^{l(\mathbf{i}) + l(\mathbf{j})}. \end{aligned}$$

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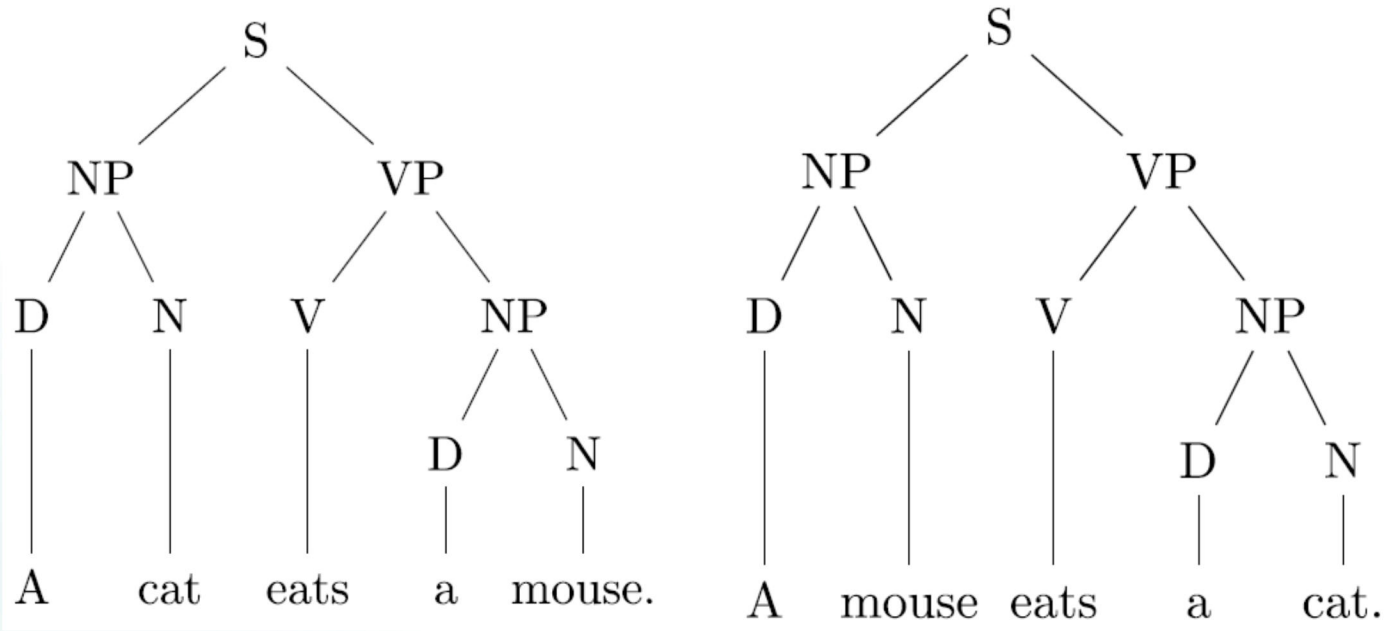
Example: Subtree kernel



- Common subtrees: [NP [D [a]] [N [cat]]], [NP [D [a]] [N [mouse]]], [N[mouse]], [N [cat]], [V [eats]], [D [a]] * 2
- Inner product = 7
- A special case of *intersection kernels*

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Curse of Dimensionality

- Quiz: Does the kernel trick make the curse of dimensionality less severe or disappear?

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Curse of Dimensionality

- From "Elements of Statistical Learning":

12.3.4 SVMs and the Curse of Dimensionality

In this section, we address the question of whether SVMs have some edge on the curse of dimensionality. Notice that in expression (12.23) we are not allowed a fully general inner product in the space of powers and products. For example, all terms of the form $2X_jX'_j$ are given equal weight, and the kernel cannot adapt itself to concentrate on subspaces. If the number of features p were large, but the class separation occurred only in the linear subspace spanned by say X_1 and X_2, this kernel would not easily find the structure and would suffer from having many dimensions to search over.

$$\begin{aligned} K(X, X') &= (1 + \langle X, X' \rangle)^2 \\ &= (1 + X_1X'_1 + X_2X'_2)^2 \\ &= 1 + 2X_1X'_1 + 2X_2X'_2 + (X_1X'_1)^2 + (X_2X'_2)^2 + 2X_1X'_1X_2X'_2. \end{aligned} \tag{12.23}$$

Curse of Dimensionality

- Example
  - Class 1: X_1, X_2, X_3, X_4 i.i.d. from standard normal
  - Class 2: X_1, X_2, X_3, X_4 i.i.d. from standard normal, condition on $9 \leq \sum X_j^2 \leq 16$
  - Noise features: $X_5, \ldots, X_{10}$ i.i.d. from standard normal
  - Training: 100 samples from each class

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Curse of Dimensionality

Method	Test Error (SE)	
	No noise feature	6 noise feature
SVM/poly 1	0.423 (0.006)	0.466 (0.008)
SVM/poly 2	0.081 (0.016)	0.172 (0.015)
SVM/poly 5	0.212 (0.008)	0.393 (0.004)
SVM/poly 10	0.265 (0.011)	0.438 (0.006)

Bayes error 0.029

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Multi-class SVMs

- Based on binary classification:
  - One-against-all
  - One-against-one
  - DAGSVM
- Consider all classes at once

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Multi-class SVMs

- One-against-all: train K one-against-all SVMs $f_1, \ldots, f_K$,

$$\hat{y} = \operatorname{argmax}_k f_k(x).$$

- One-against-one: train one SVM for every pair of classes ($K\,(K-1)/2$ SVMs in total). Predict the class by voting.

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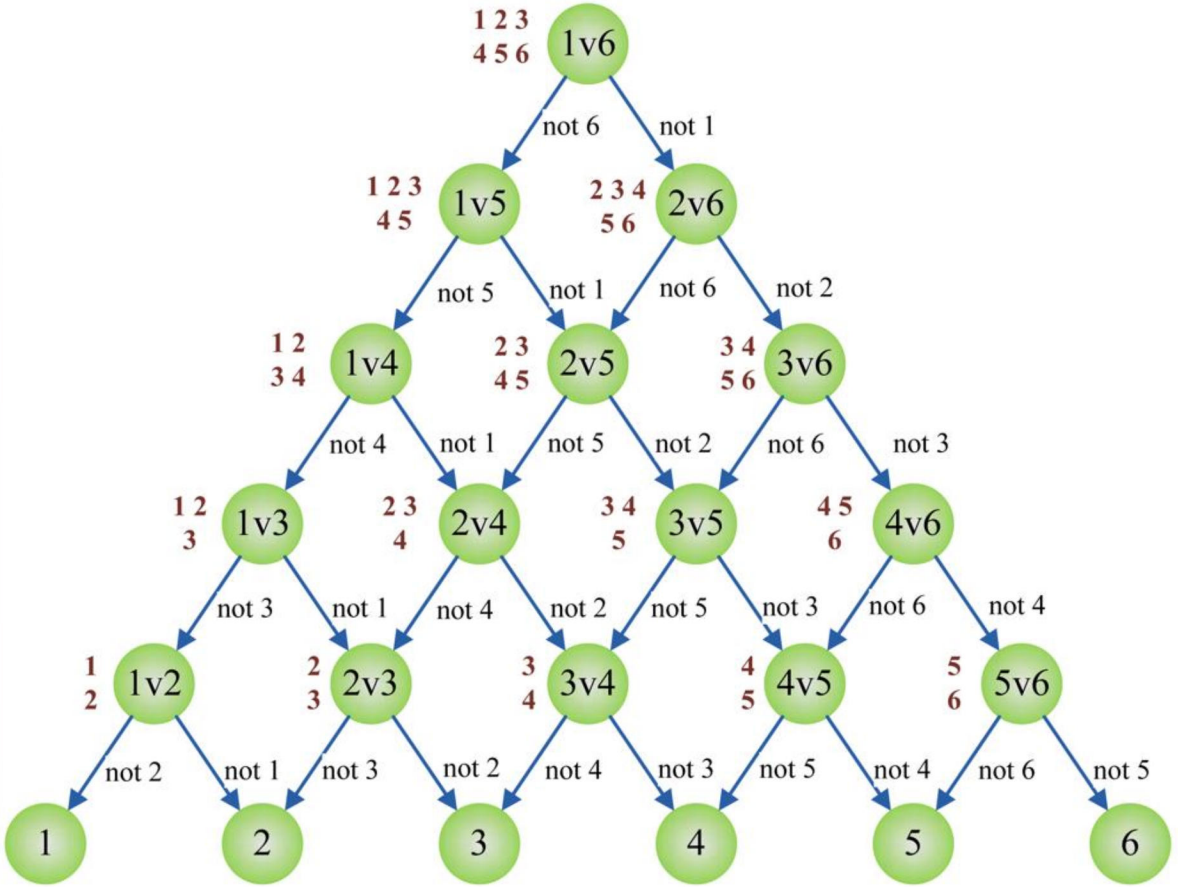
Multi-class SVMs

- DAGSVM: train one SVM for every pair of classes (still $K\,(K-1)/2$ SVMs in total), then



Multi-class SVMs

- Consider all classes at once

$$\min_{\beta,\xi} \ \frac{1}{2}\sum_{k=1}^{K}\|\beta_k\|^2 + C\sum_{i=1}^{N}\sum_{k\neq y_i}\xi_i^k$$

$$\text{s.t.} \ \ \beta_{0,y_i} + x_i^T\beta_{y_i} \geq \beta_{0,k} + x_i^T\beta_k + 2 - \xi_i^k$$

$$\xi_i^k \geq 0, i = 1, \ldots, N, k \neq y_i$$

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Advanced Topics on SVM

SVM via Loss + Penalty

- With $f(x) = h(x)^T\beta + \beta_0$,

$$\min_{\beta,\beta_0,\xi_i} \ \frac{1}{2}\|\beta\|^2 + C\sum_{i=1}^{N}\xi_i$$

$$\text{s.t.} \quad y_i f(x_i) \geq 1 - \xi_i \ \text{, for } i = 1, \ldots, N$$

$$\xi_i \geq 0 \quad \text{, for } i = 1, \ldots, N$$

is equivalent to

$$\min_{\beta,\beta_0} \ \sum_{i=1}^{N}[1 - y_i f(x_i)]_+ + \frac{\lambda}{2}\|\beta\|^2$$

with $\lambda = 1/C$.

$L(y, f) = [1 - yf]_+$ is called the "hinge" loss.

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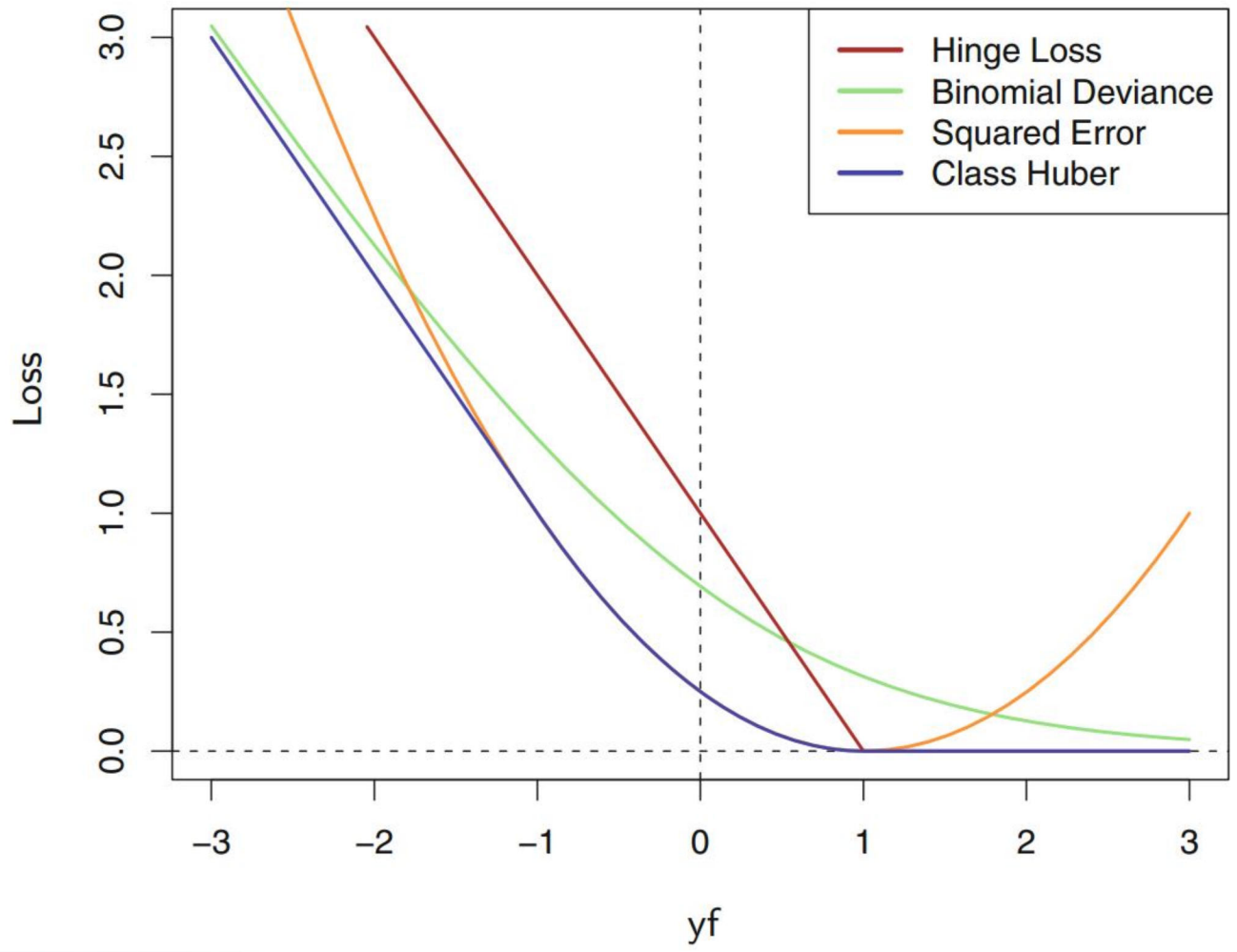
SVM via Loss + Penalty

Loss Function	$L[y, f(x)]$	Minimizing Function
Binomial Deviance	$\log[1 + e^{-yf(x)}]$	$f(x) = \log \frac{\Pr(Y = +1\|x)}{\Pr(Y = -1\|x)}$
SVM Hinge Loss	$[1 - yf(x)]_+$	$f(x) = \text{sign}[\Pr(Y = +1\|x) - \frac{1}{2}]$
Squared Error	$[y - f(x)]^2 = [1 - yf(x)]^2$	$f(x) = 2\Pr(Y = +1\|x) - 1$

SVM via Loss + Penalty



Reproducing Kernel Hilbert Space

- Like the eigenvalue decomposition of a matrix, a positive semidefinite function $K(x, x')$ has eigen expansion (Mercer's theorem)

$$K(x, x') = \sum_{j=1}^{\infty} \gamma_j \phi_j(x) \phi_j(x'),$$

where

$$\gamma_j \geq 0, \sum_{j=1}^{\infty} \gamma_j^2 < \infty;$$

$$\int \phi_i(x) \phi_j(x)\, dx = \delta_{ij}.$$

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Reproducing Kernel Hilbert Space

- In association with K, we can define an integral operator T_K, which, when applied to a function $f(x)$, generates another function:

$$T_K\big(f(x)\big) = \int K(x,y)f(y)\,dy = [T_k f](x).$$

The eigenvalues and their corresponding eigenfunctions of this operation are defined as:

$$T_K\big(\phi_i(x)\big) = \int K(x,y)\phi_i(y)\,dy = \gamma_i\phi_i(x).$$

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Reproducing Kernel Hilbert Space

- Define $\mathcal{H}_K$ to be the set of functions of the form

$$f(x) = \sum_{j=1}^{\infty} \theta_j \phi_j(x),$$

and define the inner product

$$\left\langle \sum_{j=1}^{\infty} \theta_j \phi_j(x), \sum_{j=1}^{\infty} \delta_j \phi_j(x) \right\rangle_{\mathcal{H}_K} \stackrel{\text{def}}{=} \sum_{j=1}^{\infty} \frac{\theta_j \delta_j}{\gamma_j}.$$

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Reproducing Kernel Hilbert Space

- Then the squared norm of f is

$$\|f\|_{\mathcal{H}_K}^2 = \sum_{j=1}^{\infty} \theta_j^2 / \gamma_j .$$

- $\|f\|_{\mathcal{H}_K}^2$ is generally viewed as a measurement of function *roughness*, which can be used as penalty.

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The Representer Theorem

- With $h_j(x) = \sqrt{\gamma_j}\phi_j(x)$ and $\beta_j = \theta_j / \sqrt{\gamma_j}$, $f(x) = h(x)^T\beta = \phi(x)^T\theta$, and

$$\sum_{i=1}^{N} L\big(y_i, f(x_i)\big) + \frac{\lambda}{2}\|\beta\|^2$$

$$= \sum_{i=1}^{N} L\big(y_i, f(x_i)\big) + \frac{\lambda}{2}\|f\|^2_{\mathcal{H}_K}$$

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The Representer Theorem

- The solution to the optimization problem has the finite form (Wahba 1990)

$$\hat{f}(x) = \sum_{i=1}^{N} \hat{\alpha}_i K(x, x_i)\,,$$

a finite expansion in the representers $K(x, x_i)$. If $f(x) = \beta_0 + h(x)^T\beta$, then the solution can be written as $\hat{f}(x) = \hat{\beta}_0 + \sum_{i=1}^{N} \hat{\alpha}_i K(x, x_i)$.

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The Representer Theorem

- The reproducing property: for any $f \in \mathcal{H}_K$,

$$\langle K(x, x_i), f(x) \rangle_{\mathcal{H}_K} = f(x_i).$$

Hence

$$\|\hat{f}\|^2_{\mathcal{H}_K} = \sum_{i=1}^{N} \sum_{i'=1}^{N} K(x_i, x_{i'}) \hat{\alpha}_i \hat{\alpha}_{i'} .$$

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The Representer Theorem

- Equivalent finite dimensional criterion (in matrix notation):

$$\min_{\alpha} L(y, K\alpha) + \frac{\lambda}{2}\alpha^T K\alpha,$$

where K is the $N \times N$ kernel matrix.

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Kernel Logistic Regression

- Replacing $[1 - yf]_+$ with the binomial deviance $\log\left(1 + \mathrm{e}^{-yf}\right)$, we get the kernel logistic regression.

- Quiz: Formulate the kernel logistic regression. How do you kernelize the problem? What is the classification rule?

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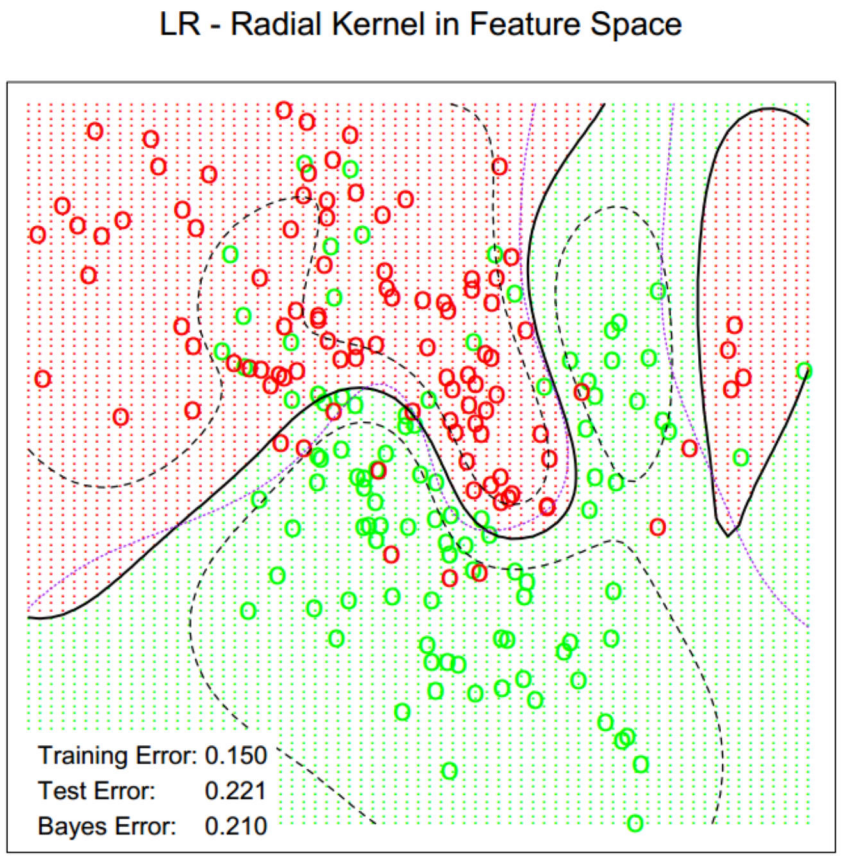
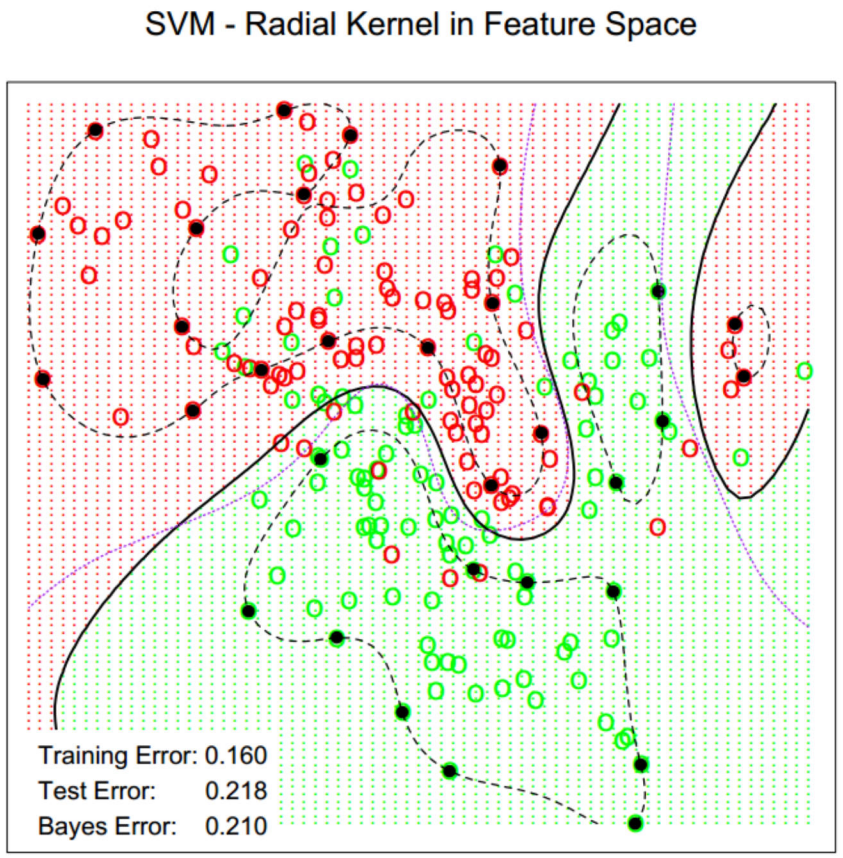
Kernel Logistic Regression

- The solid and dashed lines in the left figure are contours of $P(Y = 1|x) = 0.25, 0.5, and\ 0.75$.





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About the RBF kernel

- The RBF kernel $K(x, x') = \exp\left(-\|x - x'\|^2/\sigma^2\right)$ can be expanded as the inner product of an infinite number of features. Why doesn't it have the overfitting problem, as do the polynomial kernels?

- Since

$$K(x, x') = \sum_{j=1}^{\infty} \gamma_j \phi_j(x) \phi_j(x') \text{ and } \sum_{j=1}^{\infty} \gamma_j^2 < \infty,$$

γ_j must shrink very fast.

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About the RBF kernel

- We can apply eigen decomposition on the kernel matrix K,

$$K = Q\Sigma Q^T = \sum_{j=1}^{N} \lambda_j q_j q_j^T ,$$

and see it as an empirical approximation to $K(x, x') = \sum_{j=1}^{\infty} \gamma_j \phi_j(x)\phi_j(x')$, where the eigen values correspond to γ_j and q_j is the value of ϕ_j at various x.

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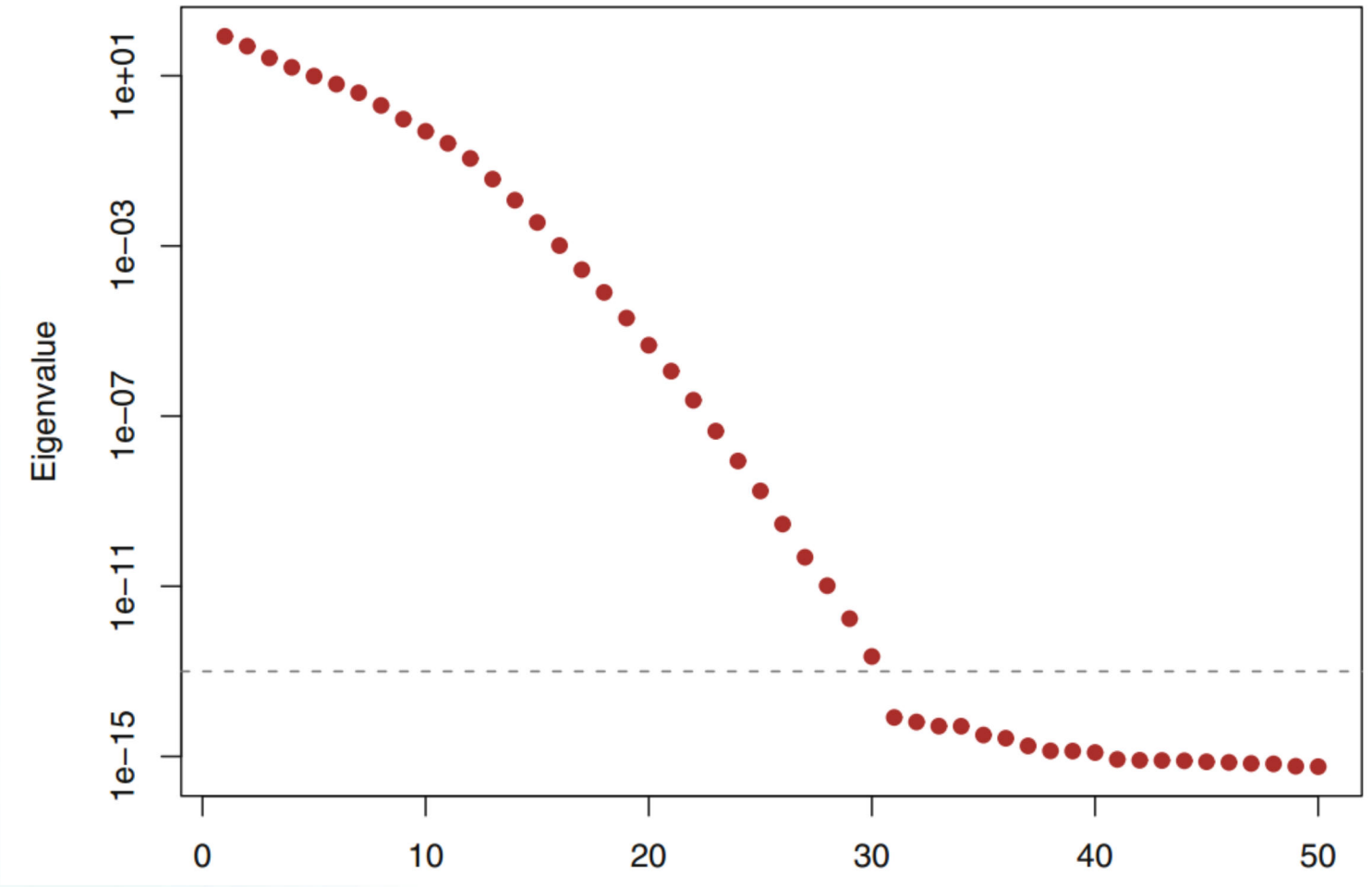
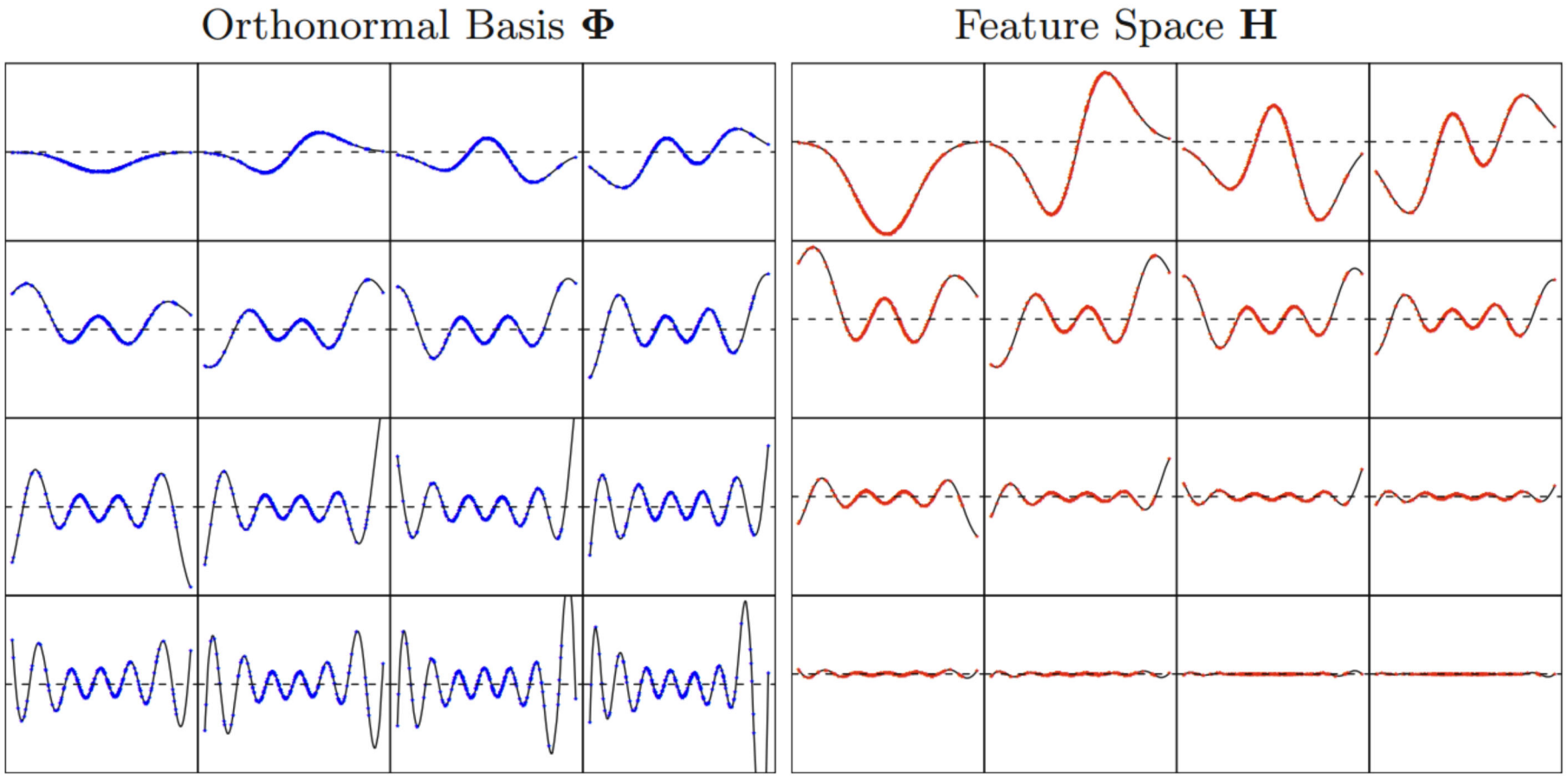
About the RBF kernel



First 50 eigenvalues from the mixture model

About the RBF kernel



Left: $\phi_j(x)$; Right: $h_j(x) = \sqrt{\gamma_j}\phi_j(x)$

About the RBF kernel

- Let $f(x) = \sum_{j=1}^{\infty} \theta_j \phi_j(x)$, and $\|\beta\|^2 = \|f\|_{\mathcal{H}_K}^2 = \sum_{j=1}^{\infty} \theta_j^2 / \gamma_j$.
- The penalty on the coefficient is magnified when γ_j is small.
- So *effectively* it is not equivalent to using an infinite number of features.

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Kernel Regression

- Let $f(x) = \sum_{j=1}^{\infty} \theta_j \phi_j(x)$. Consider the regression using squared error loss with L^2 penalty.

$$\min_{\theta_j} \sum_{i=1}^{N} \left(y_i - \sum_{j=1}^{\infty} \theta_j \phi_j(x) \right)^2 + \lambda \sum_{j=1}^{\infty} \frac{\theta_j^2}{\gamma_j}.$$

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Kernel Regression

- By the Representer Theorem, the solution has the form $\hat{f}(x) = \sum_{i=1}^{N} \hat{\alpha}_i K(x, x_i)$, and it is equivalent to solve

$$\min_{\alpha} \frac{1}{2}(y - K\alpha)^T (y - K\alpha) + \frac{\lambda}{2}\alpha^T K \alpha.$$

The solution is obtained easily as $\hat{\alpha} = (K + \lambda I)^{-1} y$.

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Kernel PCA (optional)

- Principal component analysis (PCA) is a classical dimension reduction technique for the purposes of: (1) constructing meaningful and orthogonal new variables for interpretation; (2) more stable regression; (3) visualization.

- PCA tries to find a set of p-dimensional coefficient vectors $\{w_k\}_{k=1}^{l}$ that map each sample x_i to a new vector of principal component scores $t_i = \left(x_i^T w_1, \ldots, x_i^T w_l\right)^T$ in such a way that the individual scores successively inherit the maximum possible variance from X, with each coefficient vector w constrained to be a unit vector and orthogonal to each other.

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Kernel PCA

- Assuming the columns of X have mean 0 and unit variance, the kth component can be found by subtracting the first $k-1$ principal components from X:

$$\hat{X}_k = X - \sum_{s=1}^{k-1} X w_s w_s'$$

and then finding the weight vector which extracts the maximum variance from this new data matrix

$$w_k = \arg\max_{\|w\|=1} \left\{ \left\| \hat{X}_k w \right\|^2 \right\} = \arg\max \left\{ \frac{w' \hat{X}_k' \hat{X}_k w}{w' w} \right\}.$$

It turns out that $\{w_k\}_{k=1}^{l}$ are the eigenvectors corresponding to the largest eigenvalues of the covariance matrix

$$Cov(X) = \frac{1}{N} X'X = \frac{1}{N} \sum_{i=1}^{N} x_i x_i' .$$

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Kernel PCA

- Consider a nonlinear transformation $h(x)$. We can perform standard PCA in the feature space, which implicitly defines a nonlinear principal component model in the original data space.

- For the moment, suppose $\sum_{i=1}^{N} h(x_i) = 0$.

- The sample covariance matrix in the feature space is

$$\Sigma = \frac{1}{N}\sum_{i=1}^{N} h(x_i)h(x_i)^T .$$

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Kernel PCA

- Then eigen equations become

$$\frac{1}{N}\sum_{i=1}^{N} h(x_i)\left(h(x_i)^T v_j\right) = \lambda_j v_j.$$

- Notice that v_j is a linear combination of the $h(x_i)$, so it can be written in the form

$$v_j = \sum_{i=1}^{N} \alpha_{ji} h(x_i)\,.$$

- To obtain v_j, it is sufficient to calculate α_j.

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Kernel PCA

- The eigen equations become

$$K^2\alpha_j = \lambda_j NK\alpha_j,$$

  which can be reduced to

$$K\alpha_j = \lambda_j N\alpha_j,$$

  because any solution to the second equation is a solution to the first equation.

- The normalization condition is

$$\lambda_j N\alpha_j^T\alpha_j = 1.$$

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Kernel PCA

- The resulting principal component projection is

$$z_j(x) = h(x)^T v_j$$

$$= \sum_i^N \alpha_{ji} K(x, x_i)$$

- In general, $h(x_i)$ may not have zero mean, we use

$$\widetilde{K} = K - \mathbf{1}_N K - K\mathbf{1}_N + \mathbf{1}_N K \mathbf{1}_N,$$

where $\mathbf{1}_N$ is the $N \times N$ matrix in which every element is $1/N$.

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